Object Oriented Programming Lab (CSEG1044) Experiment-15&16: Lambdas + Swing GUI (Student Handout — Before Submission)

Tofik Ali

March 3, 2026

Contents

1 About this handout **1**

2 Experiment 15 (Syllabus) **1**

3 Experiment 16 (Syllabus) **1**

4 What you must do (minimum requirements) **2**
4.1 Task A — Lambda practice (Exp-15) 2
4.2 Task B — Swing UI with events (Exp-16) 2

5 Hints (read only if stuck) **2**

6 What to submit (MANDATORY) **3**

1 About this handout

This handout specifies the required programs for lambda expressions and Swing GUI. Do the implementation yourself and include outputs/screenshots.

2 Experiment 15 (Syllabus)

- **Objective:** To use lambda expressions and functional interfaces.
- **Problem Statement:** Implement lambda expressions in Java programs.
- **Expected Outcome:** Students will understand functional programming.

3 Experiment 16 (Syllabus)

- **Objective:** To design GUI applications using Swing.
- **Problem Statement:** Develop GUI applications using Swing components.

- **Expected Outcome:** Students will create interactive applications.

4 What you must do (minimum requirements)

4.1 Task A — Lambda practice (Exp-15)

Create `LambdaDemo.java`:

1. Define a custom `@FunctionalInterface` (example: `IntOp`) with method `apply(int a, int b)`.
2. Create lambdas:
   - addition
   - max/min
3. Create a `Runnable` using lambda and run it in a thread.
4. Sort a string array using a comparator lambda (case-insensitive).

4.2 Task B — Swing UI with events (Exp-16)

Create `SwingDemo.java`:

1. Create a window with:
   - text field for name
   - button "Greet"
   - label for output
   - button "Reset"
2. On "Greet" click:
   - increment a click counter
   - update label: `Hello <name> (Clicks: X)`
3. On "Reset" click: set counter to 0 and clear the text field.
4. Use `SwingUtilities.invokeLater(...)` (EDT best practice).

5 Hints (read only if stuck)

- A lambda can be used only where a functional interface is expected.
- In Swing, create and update UI on the Event Dispatch Thread (EDT).
- Use an action listener to handle button clicks.

6 What to submit (MANDATORY)

Submit a single zipped folder containing:

- `LambdaDemo.java`
- `SwingDemo.java`
- sample output for `LambdaDemo.java`
- screenshot of the Swing UI running (or short recording)
- observations (5–10 lines): where lambdas helped; what EDT is

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed March 3, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed March 3, 2026).